

Proportional Relationships

Answer Key

Grades 6–7 ratios and proportional relationships

Quick Answers

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|----------------|---------------|-------------------|
| 1. 32, 4 | 5. 1.5, 15 | 9. 10, 0.275, 0.4 |
| 2. 48, 12, 120 | 6. 1.5, 13.5 | 10. 4, 3, — |
| 3. 2.5 | 7. 7, 8 | 11. 112.5, 7 |
| 4. 42, 20 | 8. 12, 30, 54 | 12. 3, 105, 34 |
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Curriculum

Level: Grades 6–7 ratios and proportional relationships

7.RP.A.2 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 29 tagged checks.

- 1.** Orange table: $2 \text{ kg} \rightarrow 8$, $5 \text{ kg} \rightarrow 20$, $8 \text{ kg} \rightarrow 32$, $12 \text{ kg} \rightarrow 48$ dollars. (a) Cost of 8 kg. (b) Constant of proportionality. (c) Equation. (d) Plot.

Solution (a): Read the column headed 8 straight from the table.

32

Solution (b): The constant of proportionality is $k = \frac{c}{w}$, the cost of one kilogram. Using the pair (8, 32): $k = \frac{32}{8}$. Every other column gives the same value ($\frac{8}{2} = \frac{20}{5} = \frac{48}{12}$), which is what makes the relationship proportional.

$k = 4$

Solution (c) and (d): The equation is $c = 4w$. Plotted, the four points lie on one straight line through the origin — the visual signature of a proportional relationship.

- 2.** Cyclist table: $1 \text{ h} \rightarrow 12 \text{ km}$, $3 \text{ h} \rightarrow 36$, $4 \text{ h} \rightarrow 48$, $6 \text{ h} \rightarrow 72$. (a) Distance in 4 hours. (b) Constant of proportionality. (c) Distance in 10 hours.

Solution (a): Read the table at 4 hours.

48

Solution (b): $k = \frac{d}{t} = \frac{48}{4}$, so the cyclist covers 12 km each hour.

$k = 12$

Solution (c): The equation is $d = 12t$, so at $t = 10$, $d = 12 \times 10$. Reading it off the table is impossible — 10 hours is not in the table — which is exactly why the equation is worth having.

120

- 3.** Graph of pages read against minutes. Find the unit rate and the equation.

Solution: Pick a lattice point the line clearly passes through: the marked point is (4, 10), and the line goes through the origin (0, 0). The unit rate is the slope,

$$k = \frac{\text{change in pages}}{\text{change in minutes}} = \frac{10 - 0}{4 - 0}$$

so Amir reads two and a half pages a minute and the equation is $y = 2.5x$.

$$k = 2.5$$

4. $c = 3.5n$ for n bags of soil. (a) Cost of 12 bags. (b) Bags for 70 dollars.

Solution (a): Substitute $n = 12$: $c = 3.5 \times 12$.

$$42$$

Solution (b): Now the cost is known and the count is not, so solve rather than substitute: $3.5n = 70$, so $n = 70 \div 3.5$. Check: $3.5 \times 20 = 70$ ✓

$$n = 20$$

5. Graph of litres in a tank against minutes. (a) Unit rate. (b) Litres after 10 minutes.

Solution (a): The line passes through the origin and the marked point (8, 12), so the slope is $\frac{12 - 0}{8 - 0} = 1.5$ litres a minute.

$$k = 1.5$$

Solution (b): The equation is $y = 1.5t$; at $t = 10$ minutes, $y = 1.5 \times 10$. Note that 10 minutes lies beyond the right-hand edge of the graph — the equation reaches where the drawing stops.

$$15 \text{ L}$$

6. 3 cups of flour for every 2 cups of milk. (a) Constant of proportionality. (b) Equation. (c) Flour for 9 cups of milk.

Solution (a): Flour per cup of milk is $\frac{f}{m} = \frac{3}{2}$.

$$k = 1.5$$

Solution (b) and (c): The equation is $f = \frac{3}{2}m$. With $m = 9$,

$$f = \frac{3}{2} \times 9 = \frac{27}{2}$$

so 13 and a half cups of flour.

$$f = 13.5$$

7. Table with x : 3, 5, ?, 12 and y : 21, 35, 56, 84. (a) Constant of proportionality. (b) The missing x .

Solution (a): $k = \frac{y}{x} = \frac{35}{5}$, and the other complete columns agree: $\frac{21}{3} = 7$ and $\frac{84}{12} = 7$.

$$k = 7$$

Solution (b): The missing entry must satisfy $7x = 56$, so $x = 56 \div 7$.

$$x = 8$$

8. Complete the table for $y = 6x$ at $x = 2, 5$ and 9, then plot.

Solution: Substitute each x into $y = 6x$: 6×2 , then 6×5 , then 6×9 .

$$y = 12$$

at $x = 2$;

then

$$y = 30$$

at $x = 5$;

and

$$y = 54$$

at $x = 9$.

Plotted, (2, 12), (5, 30) and (9, 54) line up with the origin, so a ruler through the origin and any one of them passes through all three.

9. Plan A: 0.25 dollars a minute. Plan B: 5 dollars plus 0.15 dollars a minute. (a) Plan A for 40 minutes. (b) Plan B cost per minute at 40 and at 20 minutes. (c) Is Plan B proportional?

Solution (a): Plan A is proportional, so multiply: 0.25×40 .

10

Solution (b): Plan B's total for 40 minutes is $5 + 0.15 \times 40 = 11$ dollars, so the cost per minute is $\frac{11}{40} = 0.275$ dollars. For 20 minutes the total is $5 + 0.15 \times 20 = 8$ dollars, so the cost per minute is $\frac{8}{20} = 0.4$ dollars.

0.275

For the shorter month the same division gives

0.4

Solution (c) — model answer (open response, review by hand): The two figures are different, and in a proportional relationship the cost per minute would have to be the same for every number of minutes — that constant is the whole point of $y = kx$. Plan B's monthly 5 dollars is charged whether you call for 20 minutes or 40, so it gets spread more thinly the more you talk, which is why the per-minute figure falls from 0.4 to 0.275. Plan B is therefore *not* proportional; its graph is a straight line, but one that crosses the vertical axis at 5 rather than passing through the origin.

10. Graph of two runners. (a) Each unit rate. (b) Compare them.

Solution (a): Both lines start at the origin, so each unit rate is just $\frac{\text{metres}}{\text{seconds}}$ at any marked point. Runner A passes through (5, 20): $\frac{20}{5} = 4$. Runner B passes through (4, 12): $\frac{12}{4} = 3$.

4

and for runner B,

3

Solution (b): The steeper line is the faster runner, and 4 metres a second beats 3 metres a second.

4 > 3

11. $d = 45t$ kilometres after t hours. (a) Distance in 2.5 hours. (b) Hours to travel 315 km.

Solution (a): Substitute $t = 2.5$: $d = 45 \times 2.5$.

112.5 km

Solution (b): Here the distance is known, so solve $45t = 315$, giving $t = 315 \div 45$, that is 7 hours. Check: $45 \times 7 = 315$ ✓

t = 7

12. Poster orders: 8 posters for 24 dollars, 20 posters for 60 dollars. (a) Constant of proportionality. (b) Equation. (c) Cost of 35. (d) Posters for 102 dollars. (e) Plot.

Solution (a): From the second order, $k = \frac{c}{n} = \frac{60}{20}$. The first order agrees: $\frac{24}{8} = 3$, which is the check that the shop really does charge proportionally.

k = 3

Solution (b) and (c): The equation is $c = 3n$, so 35 posters cost 3×35 .

105

Solution (d): Now the cost is given, so solve: $3n = 102$ gives $n = 102 \div 3$. Check: $3 \times 34 = 102$ ✓

n = 34

Solution (e): Plot (8, 24) and (20, 60); the line through them passes through the origin, and every question above can also be read off that line.